

SymPy Bug Card

Bug 59: Right Limit of Digamma at Zero Returns Zoo Instead of Negative Infinity

Task. Evaluate the right-hand limit of the order-zero polygamma function at the origin.

$$\lim_{x \rightarrow 0^+} \text{polygamma}(0, x)$$

SymPy's answer.

$$\text{limit}(\text{polygamma}(0, x), x, 0, \text{dir} = +) = \text{zoo}.$$

Correct answer.

$$\lim_{x \rightarrow 0^+} \text{polygamma}(0, x) = -\infty.$$

Explanation. Since $\text{polygamma}(0, x) = \psi(x)$, the Laurent expansion at zero is $\psi(x) = -1/x - \gamma + O(x)$. Along $x \rightarrow 0^+$, the leading term tends to $-\infty$, so `zoo` incorrectly loses the real directional sign.

Likely root cause. The diagnosis locates the omission in `polygamma._eval_as_leading_term()`, which leaves `polygamma(0, x)` unchanged instead of exposing the finite-pole leading term $-1/x$; the limit fallback then evaluates `polygamma(0, 0)` as complex infinity. Deduplication finds this duplicates an existing unmerged public fix, SymPy PR #29424 (<https://github.com/sympy/sympy/pull/29424>), which targets the same symptom and root cause.

Artifact (GitHub). [bug-report/bug-059-right-limit-of-digamma-at-zero-returns-zoo-instead-of-negati/](#)